

Corollary 2.3 *The only automorphisms of the unit disc that fix the origin are the rotations.*

Note that by the use of the mappings ψ_α , we can see that the group of automorphisms of the disc acts **transitively**, in the sense that given any pair of points α and β in the disc, there is an automorphism ψ mapping α to β . One such ψ is given by $\psi = \psi_\beta \circ \psi_\alpha$.

The explicit formulas for the automorphisms of $\mathbb{D}$ give a good description of the group $\text{Aut}(\mathbb{D})$. In fact, this group of automorphisms is “almost” isomorphic to a group of 2×2 matrices with complex entries often denoted by $\text{SU}(1, 1)$. This group consists of all 2×2 matrices that preserve the hermitian form on $\mathbb{C}^2 \times \mathbb{C}^2$ defined by

$$\langle Z, W \rangle = z_1 \bar{w}_1 - z_2 \bar{w}_2,$$

where $Z = (z_1, z_2)$ and $W = (w_1, w_2)$. For more information about this subject, we refer the reader to Problem 4.

2.2 Automorphisms of the upper half-plane

Our knowledge of the automorphisms of $\mathbb{D}$ together with the conformal map $F : \mathbb{H} \rightarrow \mathbb{D}$ found in Section 1.1 allow us to determine the group of automorphisms of $\mathbb{H}$ which we denote by $\text{Aut}(\mathbb{H})$.

Consider the map

$$\Gamma : \text{Aut}(\mathbb{D}) \rightarrow \text{Aut}(\mathbb{H})$$

given by “conjugation by F ”:

$$\Gamma(\varphi) = F^{-1} \circ \varphi \circ F.$$

It is clear that $\Gamma(\varphi)$ is an automorphism of $\mathbb{H}$ whenever φ is an automorphism of $\mathbb{D}$, and Γ is a bijection whose inverse is given by $\Gamma^{-1}(\psi) = F \circ \psi \circ F^{-1}$. In fact, we prove more, namely that Γ preserves the operations on the corresponding groups of automorphisms. Indeed, suppose that $\varphi_1, \varphi_2 \in \text{Aut}(\mathbb{D})$. Since $F \circ F^{-1}$ is the identity on $\mathbb{D}$ we find that

$$\begin{aligned} \Gamma(\varphi_1 \circ \varphi_2) &= F^{-1} \circ \varphi_1 \circ \varphi_2 \circ F \\ &= F^{-1} \circ \varphi_1 \circ F \circ F^{-1} \circ \varphi_2 \circ F \\ &= \Gamma(\varphi_1) \circ \Gamma(\varphi_2). \end{aligned}$$

The conclusion is that the two groups $\text{Aut}(\mathbb{D})$ and $\text{Aut}(\mathbb{H})$ are the same, since Γ defines an isomorphism between them. We are still left with the

task of giving a description of elements of $\text{Aut}(\mathbb{H})$. A series of calculations, which consist of pulling back the automorphisms of the disc to the upper half-plane via F , can be used to verify that $\text{Aut}(\mathbb{H})$ consists of all maps

$$z \mapsto \frac{az + b}{cz + d},$$

where a, b, c , and d are real numbers with $ad - bc = 1$. Again, a matrix group is lurking in the background. Let $\text{SL}_2(\mathbb{R})$ denote the group of all 2×2 matrices with real entries and determinant 1, namely

$$\text{SL}_2(\mathbb{R}) = \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \text{ and } \det(M) = ad - bc = 1 \right\}.$$

This group is called the **special linear group**.

Given a matrix $M \in \text{SL}_2(\mathbb{R})$ we define the mapping f_M by

$$f_M(z) = \frac{az + b}{cz + d}.$$

Theorem 2.4 *Every automorphism of $\mathbb{H}$ takes the form f_M for some $M \in \text{SL}_2(\mathbb{R})$. Conversely, every map of this form is an automorphism of $\mathbb{H}$.*

The proof consists of a sequence of steps. For brevity, we denote the group $\text{SL}_2(\mathbb{R})$ by $\mathcal{G}$.

Step 1. If $M \in \mathcal{G}$, then f_M maps $\mathbb{H}$ to itself. This is clear from the observation that

$$(4) \quad \text{Im}(f_M(z)) = \frac{(ad - bc)\text{Im}(z)}{|cz + d|^2} = \frac{\text{Im}(z)}{|cz + d|^2} > 0 \quad \text{whenever } z \in \mathbb{H}.$$

Step 2. If M and M' are two matrices in $\mathcal{G}$, then $f_M \circ f_{M'} = f_{MM'}$. This follows from a straightforward calculation, which we omit. As a consequence, we can prove the first half of the theorem. Each f_M is an automorphism because it has a holomorphic inverse $(f_M)^{-1}$, which is simply $f_{M^{-1}}$. Indeed, if I is the identity matrix, then

$$(f_M \circ f_{M^{-1}})(z) = f_{MM^{-1}}(z) = f_I(z) = z.$$

Step 3. Given any two points z and w in $\mathbb{H}$, there exists $M \in \mathcal{G}$ such that $f_M(z) = w$, and therefore $\mathcal{G}$ acts transitively on $\mathbb{H}$. To prove this,

it suffices to show that we can map any $z \in \mathbb{H}$ to i . Setting $d = 0$ in equation (4) above gives

$$\operatorname{Im}(f_M(z)) = \frac{\operatorname{Im}(z)}{|cz|^2}$$

and we may choose a real number c so that $\operatorname{Im}(f_M(z)) = 1$. Next we choose the matrix

$$M_1 = \begin{pmatrix} 0 & -c^{-1} \\ c & 0 \end{pmatrix}$$

so that $f_{M_1}(z)$ has imaginary part equal to 1. Then we translate by a matrix of the form

$$M_2 = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \quad \text{with } b \in \mathbb{R},$$

to bring $f_{M_1}(z)$ to i . Finally, the map f_M with $M = M_2 M_1$ takes z to i .

Step 4. If θ is real, then the matrix

$$M_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

belongs to $\mathcal{G}$, and if $F : \mathbb{H} \rightarrow \mathbb{D}$ denotes the standard conformal map, then $F \circ f_{M_\theta} \circ F^{-1}$ corresponds to the rotation of angle -2θ in the disc. This follows from the fact that $F \circ f_{M_\theta} = e^{-2i\theta} F(z)$, which is easily verified.

Step 5. We can now complete the proof of the theorem. We suppose f is an automorphism of $\mathbb{H}$ with $f(\beta) = i$, and consider a matrix $N \in \mathcal{G}$ such that $f_N(i) = \beta$. Then $g = f \circ f_N$ satisfies $g(i) = i$, and therefore $F \circ g \circ F^{-1}$ is an automorphism of the disc that fixes the origin. So $F \circ g \circ F^{-1}$ is a rotation, and by Step 4 there exists $\theta \in \mathbb{R}$ such that

$$F \circ g \circ F^{-1} = F \circ f_{M_\theta} \circ F^{-1}.$$

Hence $g = f_{M_\theta}$, and we conclude that $f = f_{M_\theta N^{-1}}$ which is of the desired form.

A final observation is that the group $\operatorname{Aut}(\mathbb{H})$ is not quite isomorphic with $\operatorname{SL}_2(\mathbb{R})$. The reason for this is because the two matrices M and $-M$ give rise to the same function $f_M = f_{-M}$. Therefore, if we identify the two matrices M and $-M$, then we obtain a new group $\operatorname{PSL}_2(\mathbb{R})$ called the **projective special linear group**; this group is isomorphic with $\operatorname{Aut}(\mathbb{H})$.

3 The Riemann mapping theorem

3.1 Necessary conditions and statement of the theorem

We now come to the promised cornerstone of this chapter. The basic problem is to determine conditions on an open set Ω that guarantee the existence of a conformal map $F : \Omega \rightarrow \mathbb{D}$.

A series of simple observations allow us to find necessary conditions on Ω . First, if $\Omega = \mathbb{C}$ there can be no conformal map $F : \Omega \rightarrow \mathbb{D}$, since by Liouville's theorem F would have to be a constant. Therefore, a necessary condition is to assume that $\Omega \neq \mathbb{C}$. Since $\mathbb{D}$ is connected, we must also impose the requirement that Ω be connected. There is still one more condition that is forced upon us: since $\mathbb{D}$ is simply connected, the same must be true of Ω (see Exercise 3). It is remarkable that these conditions on Ω are also sufficient to guarantee the existence of a biholomorphism from Ω to $\mathbb{D}$.

For brevity, we shall call a subset Ω of $\mathbb{C}$ **proper** if it is non-empty and not the whole of $\mathbb{C}$.

Theorem 3.1 (Riemann mapping theorem) *Suppose Ω is proper and simply connected. If $z_0 \in \Omega$, then there exists a unique conformal map $F : \Omega \rightarrow \mathbb{D}$ such that*

$$F(z_0) = 0 \quad \text{and} \quad F'(z_0) > 0.$$

Corollary 3.2 *Any two proper simply connected open subsets in $\mathbb{C}$ are conformally equivalent.*

Clearly, the corollary follows from the theorem, since we can use as an intermediate step the unit disc. Also, the uniqueness statement in the theorem is straightforward, since if F and G are conformal maps from Ω to $\mathbb{D}$ that satisfy these two conditions, then $H = F \circ G^{-1}$ is an automorphism of the disc that fixes the origin. Therefore $H(z) = e^{i\theta}z$, and since $H'(0) > 0$, we must have $e^{i\theta} = 1$, from which we conclude that $F = G$.

The rest of this section is devoted to the proof of the existence of the conformal map F . The idea of the proof is as follows. We consider all injective holomorphic maps $f : \Omega \rightarrow \mathbb{D}$ with $f(z_0) = 0$. From these we wish to choose an f so that its image fills out all of $\mathbb{D}$, and this can be achieved by making $f'(z_0)$ as large as possible. In doing this, we shall need to be able to extract f as a limit from a given sequence of functions. We turn to this point first.